

MAT135 Assignment 3 (Written Component)

Due 29 September 2024, at 9pm

Submitting the Assignment

Due date: Sunday, 29 September 2024, at 9pm.

Where to submit: On Crowdmark. Check your email for an invitation to Crowdmark.

What to do: Write solutions to the problems on blank or lined sheets of paper, or write them digitally. Start each question on a new page. Then photograph, scan, or export a PDF of your solutions. See the [Written Assignments](#) page on Quercus for details.

Uploading: Click on the link in the email from Crowdmark to submit your solutions. Plan on giving yourself some extra time to upload your solutions the first time.

More details

1. **Deadline:** The deadline to submit this assignment is strict, *to the second*. Assignments that are even a few seconds late will normally receive a grade of 0. Technical issues are not a valid reason to be late, so you are taking a risk if you leave uploading your solutions to the last minute. **Please give yourself lots of time to upload your assignment!**
2. **Purpose and feedback:** The purpose of the written assignments is to give you some practice in thinking about and writing solutions to mathematical problems, without any time pressure. You will receive feedback on your writing and on your solutions. You are encouraged to take this opportunity to carefully write your solutions and think about how to best present your reasoning behind them.
3. **Writing solutions:** *Explain all your work*, show your steps as well as your reasoning. You should write **in words** what you are doing and why. The person reading your solution should easily be able to understand what you have done, *because you explained it, in words*. Submissions with little or no written explanations will not receive full marks.
4. **Uploading:** A handwritten assignment can be photographed or scanned. It is important that the images (the scans or photos) are clear and easy to read (not too dark, too bright, too blurry, etc.). Your submitted files are what will be marked, so if the grader(s) can't read them or can't make sense of what you've written, you will not get full marks.

You may also write your solutions on a tablet, or type your solutions (LaTeX only please) as long as your solutions are clear, easy to read and follow, and are all your own work.

Make sure you upload solutions to each problem to the correct place. If you upload solutions to the wrong problems, or upload all your solutions to the same problem, it may result in getting 0 on all problems uploaded to the wrong places.

5. **Grading:** The assignment is out of 10 marks, with the marks for each question indicated beside it.

Marks for each solution will depend on your answer, as well as the quality of your explanation of those answers.
6. **Academic Integrity:** The solutions that you submit to this assignment must be all your own work. By submitting this assignment you declare that:
 - (a) Your solutions are all your own work, explained in your own words.
 - (b) You have not copied any part of the assignment solutions from anyone or anywhere.
 - (c) You have not let anyone else copy any part of your solutions to the assignment.
 - (d) You have not used Generative AI (e.g., ChatGPT) for any part of the assignment.

Background

On a peaceful day in the not-too-distant future, an enormous, mean-looking alien starship appears in the sky, scaring the people of Earth.

Using the power of trigonometry, the students of MAT135 at UTM compute the altitude of the starship by parallax. They conclude the starship is hovering in a geosynchronous orbit 420 kilometres directly above Deerfield Hall. Note that Deerfield Hall is 125 metres above sea level.

Luckily for humanity, the aliens do not know about the top-secret DH Missile Silo underneath Deerfield Hall. (You shouldn't tell anyone about it either.) This assignment is the story of what happens next.

Problems

- (1 point) The people of Earth decide to launch a Leibniz V rocket from the DH Missile Silo in an attempt to destroy the starship. Leibniz V rockets are engineered so that after launch at time $t = 0$, they fly straight up. Their altitude **above sea level** (measured in kilometres) at a time t (measured in seconds), denoted by $A(t)$, satisfies

$$A(t) = A_0 e^{pt}$$

for $t \geq 0$. The constant $p > 0$ is the **power coefficient**, which can be adjusted on each rocket as necessary.

What must the power coefficient be in order for the rocket to hit the starship in exactly 60 seconds? Give an exact answer, as well as a decimal approximation rounded to three places.

- (1 point) The rocket is launched with the power coefficient you found in Q1, but 40 seconds after launch the starship activates a temporary stun field which vaporizes any rockets below 200 kilometres above the Earth's surface.
 - (0.5 points) Verify that the first rocket was destroyed by computing the altitude of the rocket at the moment the stun field was activated. Round your answer to the nearest kilometre.
 - (0.5 points) The people of Earth decide to launch a second, faster Leibniz V to get through the stun field zone before it is activated.

What must the power coefficient be in order for it to clear the stun field at exactly 200 kilometres in altitude after 40 seconds? Call this new power coefficient P .

Give an exact value of P , as well as a decimal approximation rounded to three places.

- (1 point) The people of Earth celebrate as the second Leibniz V rocket clears the stun field. However, the instant it reaches 200 kilometres in altitude, a tractor beam appears from the bottom of the starship. It pulls the rocket up to an altitude of 400 kilometres over the next 80 seconds, then shakes it up and down for the next 3 minutes.

The altitude of the second Leibniz V at time t (measured in seconds) is observed to be $H(t)$ (again measured in kilometres):

$$H(t) = \begin{cases} A_0 e^{Pt} & 0 \leq t < 40 \\ at + b & 40 \leq t < 120 \\ 5 \cos(\pi(t - 120)) + 395 & 120 \leq t \leq 300 \end{cases}$$

Find the values of a and b , and explain how you got your answers.

Scientists are eager to find out what happened, and want to understand more about H .

- Let h be the restriction of H to the time interval before the shaking started. In other words, $h(x) = H(x)$, but h is only defined on $[0, 120]$.
- Also let g be the restriction of H to $[0, 121]$.

4. (3 points)

- (1 point) What are the ranges of h and g ? Explain your answer.
- (1 point) Explain (in a few sentences) how we can tell that h is invertible.
- (1 point) Is g invertible? Explain your answer.

5. (2 points) Find the (piecewise) formula for h^{-1} .

6. (2 points) After 5 minutes ($t = 300$), the tractor beam flings the Leibniz V rocket into deep space never to be seen again, then begins slowly descending to Earth. The starship lands in the field at the centre of the UTM campus, where it fits perfectly, at time $t = 600$.

During this period of time, the starship's altitude (in kilometres) above sea level at a time $t \geq 300$ (in seconds) is given by $S(t)$.

- (1 point) The function S is not known to humanity, but explain in one or two sentences how an observer watching the starship land could visually confirm whether S is invertible or not on its domain of $[300, 600]$.
- (1 point) Except for dealing with the Leibniz V rockets, the alien starship did not act aggressively or hurt anyone. However, people who watched the starship land observed a fly buzzing around in the middle of the field, and they do not believe it got away in time.

If $F(t)$ represents the altitude (in kilometres) of this unfortunate fly at time t (in seconds), compute $\lim_{t \rightarrow 600^-} S(t)$ and $\lim_{t \rightarrow 600^-} F(t)$, and justify your answers. Your justification should involve using a theorem we discussed in class.